

Transformative Technology:
The Role of Artificial Intelligence in the Medical and Healthcare Fields

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Rhetoric 1

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Exordium: According to the Journal of the American Medical Association, the healthcare system in the United States wastes as much as “\$935 billion, accounting for approximately 25% of total health care spending”¹ each year. If the healthcare system were to redirect even a fraction of this excessive waste to help improve patient care, the effect would be enormous. Now, imagine a world where this improved patient care means that diseases do not simply appear without warning but are predicted and diagnosed before they even occur. A world where treatment plans are not just simple and generic but are detailed with precision to each patient’s needs. Fortunately, the power of artificial intelligence (AI) can turn this dream into a reality.

Narratio: AI is a branch of computer science in which computer algorithms perform human-like tasks.² In the 1950s, Alan Turing became one of the first researchers associated with the concept of AI. His paper, “Computing Machinery and Intelligence,” explored the possibility of machine intelligence.³ In 1956, the first AI program, known as Logic Theorist, was presented at the *Dartmouth Summer Research Project on Artificial Intelligence*. Not only was the first AI program presented at this conference, but the term, Artificial Intelligence, was first coined here as well by one of its hosts, John McCarthy.⁴ Since then, AI has continued to grow but has been largely dependent upon funding, public interest, and computer capabilities. Due to recent interest in AI, more technology companies are developing AI algorithms in an effort to increase

¹ William H. Shrank, Teresa L. Rogstad, Natasha Parekh. “Waste in the US Health Care System: Estimated Costs and Potential for Savings,” *JAMA* 322, no. 15 (October 2019): 1501-1509.

² Jianxing He, Sally L. Baxter, Jie Xu, Jiming Xu, Xingtao Zhou, and Kang Zhang, “The practical implementation of artificial intelligence technologies in medicine,” *Nature Medicine* 25, no. 1 (January 2019): 31.

³ Rockwell Anyoha, “The History of Artificial Intelligence,” *Science in the News*, August 28, 2017, under “Can Machines Think?”

⁴ Rockwell Anyoha, “The History of Artificial Intelligence,” under “The Conference that Started it All.”

computational speeds. To predict how computer capabilities improve over time, Gordon Moore created Moore's Law. Moore originally predicted that computer algorithms would double in computational speed every 20 months. However, "Since about 2010, this exponential growth has sped up further, to a doubling time of just about 6 months."⁵ With this exponential growth, AI has found applications in various sectors, including business, vehicles, and everyday technology, such as your phone. But perhaps one of the most promising areas for AI is in medicine and healthcare.

In today's society, the demands on the healthcare and medical fields are higher than ever. Despite the US spending more on healthcare than any other country in the world,⁶ Americans are increasingly unhealthy. According to the U.S. Department of Health and Human Services, "Six of 10 Americans have at least one chronic condition and 4 of 10 have two or more chronic conditions."⁷ Furthermore, with almost 17% of the US population being 65 and older, the demand for healthcare services continues to increase.⁸ The cost of healthcare is also rising, putting financial strain on both patients and healthcare providers. As the New England Journal of Medicine summarizes, "Health care today is often characterized by mediocre quality, poor safety, and high costs."⁹ This is why, in response to the demands of today's healthcare system, artificial intelligence should be implemented in medicine and healthcare.

⁵ *Our World in Data*, December 6, 2022, under "What is next?"

⁶ William H. Shrank, "Waste in the US Health Care System," 1501-1509.

⁷ U.S. Department of Health and Human Services, "Community Health and Economic Prosperity The Problem, the Causes, the Opportunities, and the Solutions—At a Glance," *U.S. Department of Health and Human Services*, January 2021, 3.

⁸ Caplan, Zoe, "2020 Census: 1 in 6 People in the United States Were 65 and Over," *United States Census Bureau*, March 25, 2023. <https://www.census.gov/library/stories/2023/05/2020-census-united-states-older-population-grew.html>.

⁹ Eyal Zimlichman, Wendy Nicklin, Rajesh Aggarwal, David W. Bates, "Health Care 2030: The Coming Transformation," *NEJM Catalyst*, (March 2021): 1.

Partitio: I will argue that AI should be used in medicine and healthcare by showing how it can improve operational efficiency and reduce physician burnout, effectively restoring physician-patient trust. I will then present three counterarguments against AI in medicine and healthcare and show possible solutions. Finally, I will present instances where AI has successfully predicted, diagnosed, and treated diseases.

Confirmatio I: First, AI has the potential to improve operational efficiency within the medical and healthcare fields, including saving time, resources, and money. As mentioned earlier, medicine and healthcare are often associated with high costs. To put this into perspective, a staggering 20 percent of all healthcare spending goes just to administrative costs.¹⁰ These administrative costs include appointment scheduling, insurance claims, filing, reviewing, or resolving accounts.¹¹

With its fast, cost-effective tools, AI is a game-changer in streamlining administrative tasks. For instance, appointment scheduling, which usually requires human intervention, can be replaced with natural-language processing (NLP), which is a subfield of AI. NLP uses technology to help computers understand human language and can significantly reduce the need for human interaction since it is available around the clock.¹² Possibly the most important use for AI in administrative tasks is dealing with insurance. AI can analyze large amounts of data to provide personalized insurance plans and allow hospitals to accept a variety of plans.¹³ This could lead to more accurate pricing, better risk management for insurance companies, and more

¹⁰ Eric, Topol. “Deep Medicine: How Artificial Intelligence Can Make Healthcare Human Again,” New York, NY: Basic Books, 2019, 193.

¹¹ Drexel University. 2021. “Pros & Cons of Artificial Intelligence in Medicine,” *Drexel University*, July 21, 2021, under “Saves Time and Resources.”

¹² IBM. n.d. “What is natural language processing (NLP)?,” *IBM*, under “What is NLP?”

¹³ Drexel University, “Pros and Cons of Artificial Intelligence,” under “Streamline Tasks.”

cost-effective plans for patients. Ultimately, using AI for administrative tasks results in better cost savings and efficiency.

Another way in which AI improves operational efficiency is during clinical trials. In medicine, clinical trials are a crucial process in developing necessary drugs and medicine. However, this process is very time-consuming and costly. Fortunately, AI can speed up the clinical trial process at a much lower cost. One way that AI is more efficient is by assigning medical codes. In clinical trials, “Medical coding is performed to categorize the medical terms reported appropriately so that they can be analyzed/reviewed.”¹⁴ Recently, IBM’s AI algorithm helped reduce medical code searches by 70%.¹⁵ AI can also perform quick statistical analyses on data and review text or write clinical trial summaries, all contributing to better efficiency and cost.¹⁶ Furthermore, AI can assist in the development of drugs for clinical trials, by “creating better drug designs and finding promising new drug combinations.”¹⁷ Research companies are becoming more dependent upon AI and by 2018, “more than sixty start-ups and sixteen pharmaceutical companies were using AI for drug discovery in clinical trials.”¹⁸ One such company, DeepMind, has developed an AI system that assists in understanding disease processes and developing targeted treatments. This AI system “provides [scientists with] open access to over 200 million protein structure predictions to accelerate scientific research.”¹⁹ Having access

¹⁴ Babre, Deven, “Medical Coding in Clinical Trials,” *Perspectives in Clinical Research* 1, no. 1 (Jan 2010): 1.

¹⁵ IBM. n.d. “What is artificial intelligence in medicine?,” *IBM*, under “Clinical Trial Efficiency.”

¹⁶ Victor Grech, Sarah Cuschieri, Abdelazeem A. Eldawlatly, “Artificial intelligence in medicine and research – the good, the bad, and the ugly,” *Saudi Journal of Anesthesia* 17, no. 3 (July-September 2023): 402.

¹⁷ IBM, “What is artificial intelligence in medicine?,” Accelerated Drug Development.

¹⁸ Eric Topol, “Deep Medicine,” 215.

¹⁹ Alpha Fold, “AlphaFold Protein Structure Database,” *Google DeepMind and EMBL-EBI*, 2024.

to a database with all known protein structures helps scientists speed up scientific discoveries, including drug development, disease understanding, and bioengineering.

A third way in which AI improves operational efficiency within the medical and healthcare field is in the interpretation of medical scans like chest X-rays. AI is now capable of achieving a 95% accuracy rate in chest X-rays, which in some cases is more precise than a human's interpretation.²⁰ AI's precision is due to its ability to observe details in every pixel of the image that the human eye cannot see.²¹ This is especially important considering that medical error is the third leading cause of death in the US.²² With AI's precision in interpreting medical scans, we could see a significant reduction in medical errors associated with medical scan diagnoses. Zebra Medical Vision's AI is a prime example of this technological advancement in healthcare. This AI is "capable of reading 260 million scans in just twenty-four hours for only \$1,000," whereas many doctors, such as radiologists, can only observe 20,000 medical scans per year.²³ As AI gains access to millions of new scans each year, its accuracy and ability to interpret these scans will improve, which will greatly benefit the medical field.

Confirmatio II: With increasing demands on the healthcare system, AI can prevent physician burnout and stress in healthcare while also restoring the lost bond between physicians and patients. In the healthcare field, one resource that physicians want back most is their time. Many physicians experience stress due to tight deadlines and work conditions which can eventually lead to a condition known as physician burnout. Physician burnout is characterized by

²⁰ Eric Topol, "Deep Medicine," 116.

²¹ Ibid.

²² Rutkin, Aviva, "Medicine by machine," *New Scientist* 231, no. 3089 (September 2016): 20.

²³ Eric Topol, "Deep Medicine," 115-120,.

exhaustion, apathy towards patients, and depression.²⁴ Time constraints and stressful work conditions have contributed to a significant decline in the quality of physician-patient relationships.

AI can help reduce physician stress by taking over a doctor's daily tasks, thus reducing medical errors. In the high-pressure world of healthcare, one study identified that “nearly 63% of physicians [are] reporting signs of burnout such as emotional exhaustion and depersonalization at least once per week.”²⁵ As Eric Topol explains, “Burnout leads to medical errors, and medical errors in turn promote burnout.”²⁶ AI can already interpret medical scans with impressive speed and accuracy, which would allow doctors to spend less time interpreting scans themselves, and even provide information about the scan that doctors may not notice. Furthermore, AI can aid in accurate diagnosis, thus preventing the harmful effects of misdiagnosis. Misdiagnosis, as stated earlier, is the third leading cause of death in the United States.²⁷ By utilizing AI, valuable time can be returned to doctors, alleviating a significant amount of stress they experience from their daily work requirements. This not only increases the well-being of doctors but also reduces the risk of medical errors, increasing the safety of healthcare for all.

Although AI can improve time management and reduce possible medical errors, some express concerns that an over-dependence on AI could decrease learning opportunities for doctors and hinder their clinical judgment. However, this argument is not valid for three

²⁴ American Medical Association, “What is physician burnout?,” *American Medical Association*, February 16, 2023, under “Contents.”

²⁵ American Medical Association, “What is physician burnout?,” under “What experts are saying about burnout.”

²⁶ Eric Topol, “Deep Medicine,” 285-286.

²⁷ Rutkin, Aviva, “Medicine by machine,” 20.

reasons.²⁸ First, AI in healthcare is designed to assist, not replace doctors. AI can identify areas where doctors could improve, providing them with feedback on their decisions. If a doctor makes a diagnosis that the AI system identifies as potentially incorrect, it can provide the doctor feedback, allowing them to learn and improve, without putting the patient at risk of a misdiagnosis. Secondly, research has shown a “synergistic effect” when doctors and AI work together.²⁹ AI can serve as a triage tool by analyzing medical images and determining which patients should be seen first based on the urgency of the case at hand.³⁰ This does not lead to a decline in doctor competence, but rather creates a more efficient healthcare team. Lastly, the human element in healthcare, especially in terms of emotions and rational thinking, cannot be ignored. Even though reliance on AI might reduce the number of tasks doctors perform, the human aspect of care remains invaluable. In an AAMC Public Opinion Research poll, researchers found that patients value a doctor who “treats the whole person not just the specific medical condition” and one who “listens and takes time to answer questions.”³¹ Although their workload may change, doctors still play a critical role in patient healthcare.

While it is clear that patients value a strong relationship with their doctors, the demands of today’s healthcare system can negatively impact the physician-patient relationship.

Fortunately, AI can share workloads, allowing doctors more time with their patients, effectively restoring the physician-patient bond. For example, AI can give patients and doctors access to

²⁸ Anishchenko, Mykhailo A., Ievgen Gidenko, Maksym Kaliman, Vasyl Polyvaniuk, and Yurii V. Demianchuk, “Artificial Intelligence in Medicine: Legal, Ethical and Social Aspects,” *Acta Bioethica* 29, no. 1 (June 2023), 69.

²⁹ Jianxing He, Sally L. Baxter, Jie Xu, Jiming Xu, Xingtao Zhou, and Kang Zhang, “The practical implementation of artificial intelligence technologies in medicine,” *Nature Medicine* 25, no. 1 (January 2019), 30.

³⁰ *Ibid.*

³¹ Redford, Gabrielle, “What makes a good doctor — and other findings from the 2019 AAMC Public Opinion Research,” *AAMC*, April 27, 2020.

real-time data. Patients can access their own data through mobile apps so they can better engage in treatments. By using these apps, “Mobile alerts can inform doctors and nurses of urgent changes in patient statuses and emergencies.”³² This interactive real-time data can help both patients and doctors reestablish trust. AI technologies that utilize natural language processing (NLP) are also capable of automating data collection, allowing doctors more time to focus on their patients. For instance, Emerald’s touchless sensor and Google Nest’s sleep monitoring system actively monitor a patient’s health remotely.³³ Doctors can then use these metrics to quickly assess a patient’s health. This technology provides doctors with expert knowledge that helps them treat patients more efficiently, reestablishing this lost bond.

Refutatio: Despite its many benefits, some believe that AI should not be implemented in medicine and healthcare altogether. They argue that continuously updating AI algorithms with new data is difficult because of privacy concerns.³⁴ This data can include medical histories, genetic information, lifestyle habits, and more. The concern is that if this information fell into the wrong hands or was misused, this would be considered a breach of privacy. AI does, in fact, require large amounts of diverse and comprehensive data for training to ensure accurate medical predictions. However, patient information can be protected. One solution is to give patients control over their data, through the use of consent forms that limit who can access it and for what purposes. Luckily, regulations can also help protect patients. The General Data Protection and Regulation (GDPR) is one such regulation that “requires explicit and informed consent before

³² Drexel University, “Pros and Cons of Artificial Intelligence,” under “Provides Real-Time Data.”

³³ Junaid Bajwa, Usman Munir, Aditya Nori, Bryan Williams, “Artificial intelligence in healthcare: transforming the practice of medicine,” *Future Healthcare Journal* 8, no. 2 (July 2021), 191.

³⁴ He et al., “Practical Implementation of Artificial Intelligence,” 31.

any collection of personal data.”³⁵ Ultimately, consent forms and data regulation will help ensure that AI algorithms are continuously updated while protecting a patient’s personal information.

Another major argument is that if an AI system provides a diagnosis resulting in harm or death, some believe that it would be impossible to determine who is liable.³⁶ Clear regulations and guidelines can help prevent such situations. This could include specialized testing and certification of AI systems before use in healthcare settings. To further ensure patient safety, doctors could provide human oversight in order to review AI-generated diagnoses before finalizing a treatment plan.

Another argument against AI is that algorithms could be trained with biased data.³⁷ Bias could lead to unfair healthcare outcomes where some groups of people get better care than others. To reduce the possibility of bias, AI systems must be trained with diverse sets of data. Diverse data can include variables such as age, gender, socioeconomic status, race, ethnicity, and even location. AI algorithms must also be continuously validated and improved to avoid bias. Furthermore, it is crucial to design AI algorithms with transparency. A common concern regarding transparency is the so-called ‘black box’ problem, which refers to the inability to see how AI algorithms reach a certain conclusion.³⁸ A solution to this is designing AI algorithms so that they provide clear explanations for their decisions, thus aiding in transparency. Doing so will help physicians understand how the algorithm reached its conclusion and help identify any potential biases.

³⁵ He et al., “Practical Implementation of Artificial Intelligence,” 34.

³⁶ Anishchenko et al., “Legal, Ethical and Social Aspects,” 65-66.

³⁷ Drexel University, “Pros and Cons of Artificial Intelligence,” under “May Overlook Social Variables.”

³⁸ Blouin, Lou, “AI’s mysterious ‘black box’ problem, explained,” *University of Michigan-Dearborn*, March 6, 2023.

Confirmatio III: Despite these concerns, AI is already being rigorously tested and has shown great potential in the healthcare and medical fields. In terms of predictive AI, AI is being used to predict diseases before symptoms appear. One company called Medtronic has created an AI system called Guardian Connect that tracks blood sugar levels and transmits the gathered information to a mobile app.³⁹ Guardian Connect uses IBM's AI technology "to predict major fluctuations in blood, glucose levels 10-60 minutes in advance."⁴⁰ In this case, Medtronic's Guardian Connect monitors blood sugar levels and serves as a reliable and predictive tool for people with diabetes. On the other hand, an AI system created by Aidoc, called BriefCase, can detect diseases by identifying abnormal patterns in the data.⁴¹ Briefcase acts as a triage tool to alert doctors of suspected "time-sensitive conditions...to be prioritized for reading."⁴² This predictive AI can lead to faster diagnoses and treatments for patients. Researchers in Los Angeles have presented AI that can predict the development of kidney and heart disease.⁴³ Ultimately, AI's remarkable ability to predict diseases before they occur is significantly changing the medical and healthcare fields. By observing large amounts of data, AI can predict diseases before symptoms even appear, allowing doctors to intervene early and provide personalized care.

In addition to predicting diseases, AI is also being used to diagnose patients with diseases. For instance, Google trained an AI system from over 100,000 retinal images from 75,000 patients to diagnose diabetic retinopathy.⁴⁴ According to Mayo Clinic, diabetic

³⁹ He et al., "Practical Implementation of Artificial Intelligence," 34.

⁴⁰ Ibid.

⁴¹ Lyell, David, Farah Magrabi, "Artificial intelligence in medicine: has the time come to hang up the stethoscope?," *Internal Medicine Journal* 53, no. 9 (September 2023), 1535-1536.

⁴² Ibid.

⁴³ Rutkin, Aviva. "Medicine by machine," 20.

⁴⁴ Eric Topol, "Deep Medicine," 145-146.

retinopathy “is a diabetes complication that affects the eyes. It’s caused by damage to the blood vessels of the light-sensitive tissue at the back of the eye.”⁴⁵ A board of more than sixty ophthalmologists reviewed these images and found the AI to have “an impressive sensitivity of 87 to 90 percent and 98 percent specificity.”⁴⁶ One real-world example of AI in action is IBM’s Watson. This AI-powered supercomputer once successfully saved a woman’s life by looking at her cancer symptoms and diagnosing her with a rare form of secondary leukemia.⁴⁷ A disease that had been perplexing her doctors for months was easily solved within a few minutes thanks to Watson.⁴⁸ If doctors had not done this, it might have been too late to save the woman’s life. Similarly, AI is being implemented to assist The Kashi First People’s Hospital in a mountainous area in China with a population of 4.5 million people. Their AI system employs retinal photographs to screen for and diagnose many common diseases here, including eye diseases, diabetes, hypertension, and stroke.⁴⁹

Finally, AI is being more widely used to treat patients. One major advantage to AI is that it can give patients more personalized treatment since it can “remember personal preferences and [is] available round the clock.”⁵⁰ AI algorithms can analyze large amounts of a patient’s data to create the most effective treatment possible. AI can analyze factors such as DNA, lifestyle, and health history. This personalized approach can significantly reduce the number of trial-and-error approaches by aiming at a correct treatment the first time. In addition to personalized treatments,

⁴⁵ Mayo Clinic, “Diabetic retinopathy,” *Mayo Clinic*, 2024.

⁴⁶ Eric Topol, “Deep Medicine,” 145-146.

⁴⁷ Rutkin, Aviva. “Medicine by machine,” 20.

⁴⁸ Ibid.

⁴⁹ He et al., “Practical Implementation of Artificial Intelligence,” 34.

⁵⁰ Victor Grech, Sarah Cuschieri, Abdelazeem A. Eldawlatly. “The good, the bad, and the ugly,” 402.

AI surgical robots have gained popularity over the past few decades due to their cost-effectiveness and precision. The most common surgical robot performing these operations is the da Vinci surgical system. As explained by Eric Topol, “In 2016 alone, more than 4,000 such robots helped perform 750,000 operations around the world.”⁵¹ Although surgical robots perform less than 10% of surgeries, the growing popularity of AI will ensure their continued success.⁵²

Peroratio: With all these advantages, it’s clear that the future of medicine and healthcare includes AI. The integration of AI in medicine and healthcare offers potential savings in time, resources, and money. AI can also alleviate physician burnout, restoring the physician-patient relationship. Finally, AI has already demonstrated its ability to predict, diagnose, and treat diseases in the medical field.

Recognizing the significance of AI is crucial for anyone involved in or affected by the medical and healthcare fields. By improving operational efficiency and reducing physician burnout, AI can enhance the quality of care for patients and decrease the number of misdiagnoses. It also helps restore the vital bond between physicians and patients. Furthermore, the fact that AI has already shown great potential to predict, diagnose, and treat diseases in these fields underscores the promising possibilities for the future.

From a practical standpoint, it is important to support policies that encourage the use of AI in healthcare while safeguarding patient information. Not only can AI save the medical and healthcare fields both time and money, it can do the same for patients. By identifying diseases in advance, patients can save trips to the doctor’s office, money on prescriptions, and improve their overall health and well-being. Ultimately, AI is here to stay, and the sooner that patients embrace this technology in the medical and healthcare fields, the sooner they can improve their lives.

⁵¹ Eric Topol, *Deep Medicine*, 160-161.

⁵² Ibid.

One vital question remains. As Christians, should we embrace AI in medicine and healthcare? The arrival of AI in medicine and healthcare presents several tough issues. An important consideration is that “At the heart of all this, human beings are creating the algorithms that drive AI. As such, AI reflects all the good and bad that human beings are capable of.”⁵³ Fortunately, AI algorithms continue to advance, becoming more and more reliable. AI’s benefits cannot be understated. Its ability to save time, resources, and money is vital in helping the healthcare system. Similarly, AI can significantly reduce the doctor’s workload, offering a solution to the important issue of physician burnout. By taking over simple or time-consuming tasks, AI allows physicians to focus more on their patients instead of less important tasks, creating a stronger physician-patient relationship. This increased interaction emphasizes an important biblical idea - the need for human interaction. Humans are not meant to go through life’s challenges alone, and the use of AI in healthcare not only improves operational efficiency but also demonstrates the need for relationships between doctors and their patients. Furthermore, the Bible teaches that human life is invaluable and should be protected. AI has the potential to prevent misdiagnosis, which aligns with this idea of saving human life. It has already shown its ability to predict, diagnose, and treat diseases, demonstrating its potential to serve humanity. As Rajan Mathews explains, “No one living in our modern world can escape the impact of AI. So, as Christians, we must not only get familiar with its use but also gain proficiency to manage and even design it for the church and the common good.”⁵⁴ In conclusion, despite AI being created by sinful humans, Christians should view AI in medicine and healthcare as a tool that has the potential to do good. Ultimately, the future of healthcare is here, and AI powers it.

⁵³ Rajan S. Mathews, “How Should Christians Think About Artificial Intelligence?,” under “What is Artificial Intelligence?”

⁵⁴ Rajan S. Mathews, “How Should Christians Think About Artificial Intelligence?,” under “Actively Take Part in Shaping the Future of AI.”

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